Google Cloud H2S

AI Builder Cup

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Team Details

TwinOps: a rehearsal space for network repairs

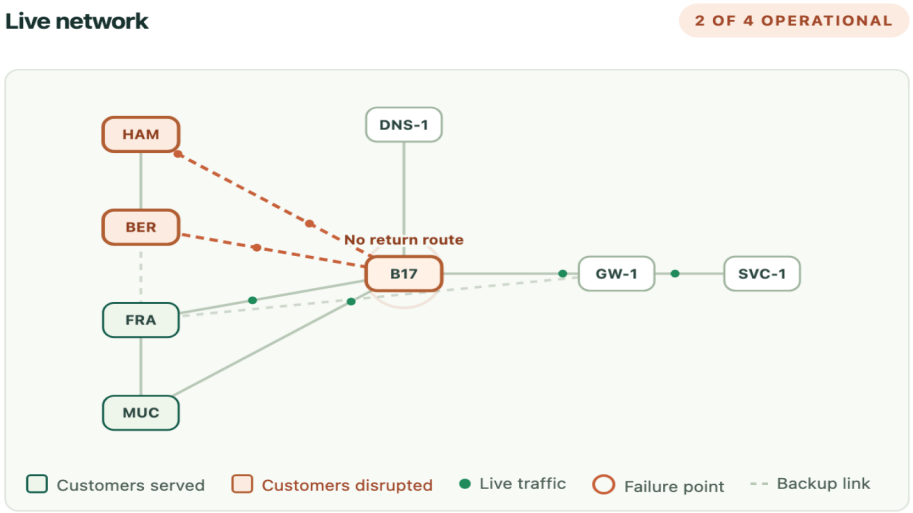
AI tests a fix in a separate network. An engineer decides what to apply.

HackSmith / Prabhakar Srivastava (leader) / Shivam Anand

Theme: Future of Work & Enterprise Productivity

Brief about the idea

Two branch sites lose access. Which repair can we trust?



The engineer's problem

A quick rollback might restore access while breaking another connection.

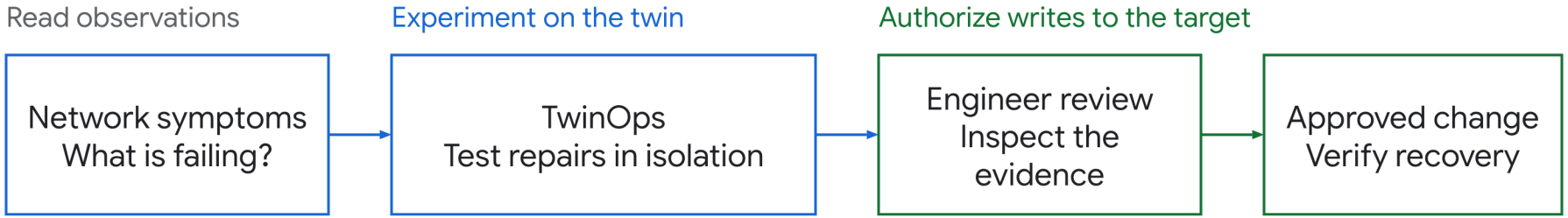
TwinOps rehearses the repair

A digital twin is a separate copy of the affected network. AI tests alternatives there and gives the engineer evidence before asking for approval.

Metro is our synthetic application network. A "journey" is a site-to-service connectivity check.

Opportunities

TwinOps sits between an incident and an approved change.



Different approach Test competing repairs and challenge them before change review.

Problem solved A fix that restores access can still fail another required check.

Unique value One reviewable record links the exact change, tests and rollback.

List of features offered by the solution

TwinOps explains why one repair survives and another fails.

Gemini proposes repairs. Twin tests and fixed policy checks decide which can proceed.

Recorded candidate	Access restored	IPv6 check	Decision
Remove one bad route rule	Yes	Pass	Passed tests
Roll back the full configuration	Yes	Fail	Stopped

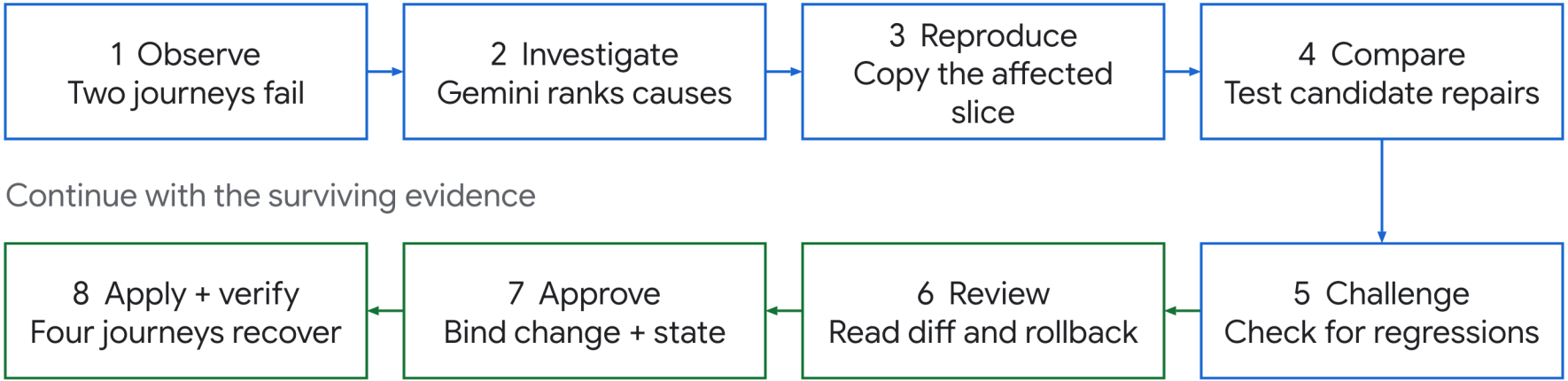
Both restored the initial journeys. Only the narrow repair preserved IPv6 reachability.

The workbench retains rejected alternatives, the exact diff and a rehearsed rollback for review.

Process flow diagram

One routing incident, from failed access to verified recovery

Steps 1–5: TwinOps reads the application and experiments only on its twin.

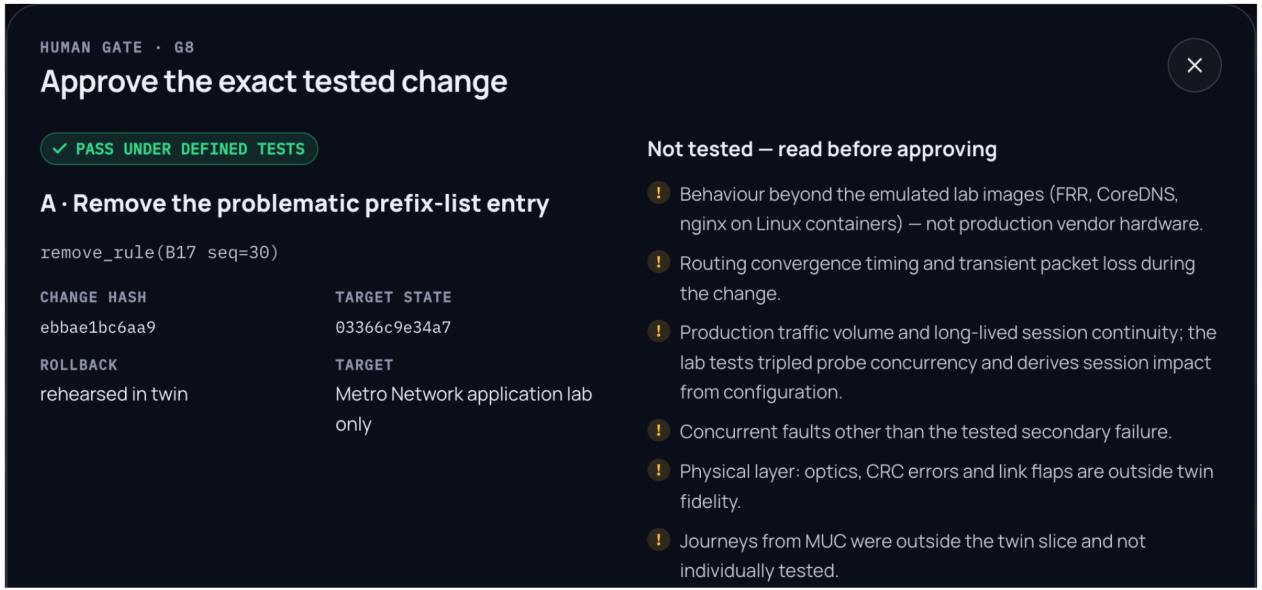


Continue with the surviving evidence

Missing evidence blocks a proposal. Failed recovery triggers rollback handling and escalation.

Wireframes / interface: the implemented workbench

The engineer approves a specific, tested change.



Exact change

Remove rule 30 on router B17.

Tested rollback

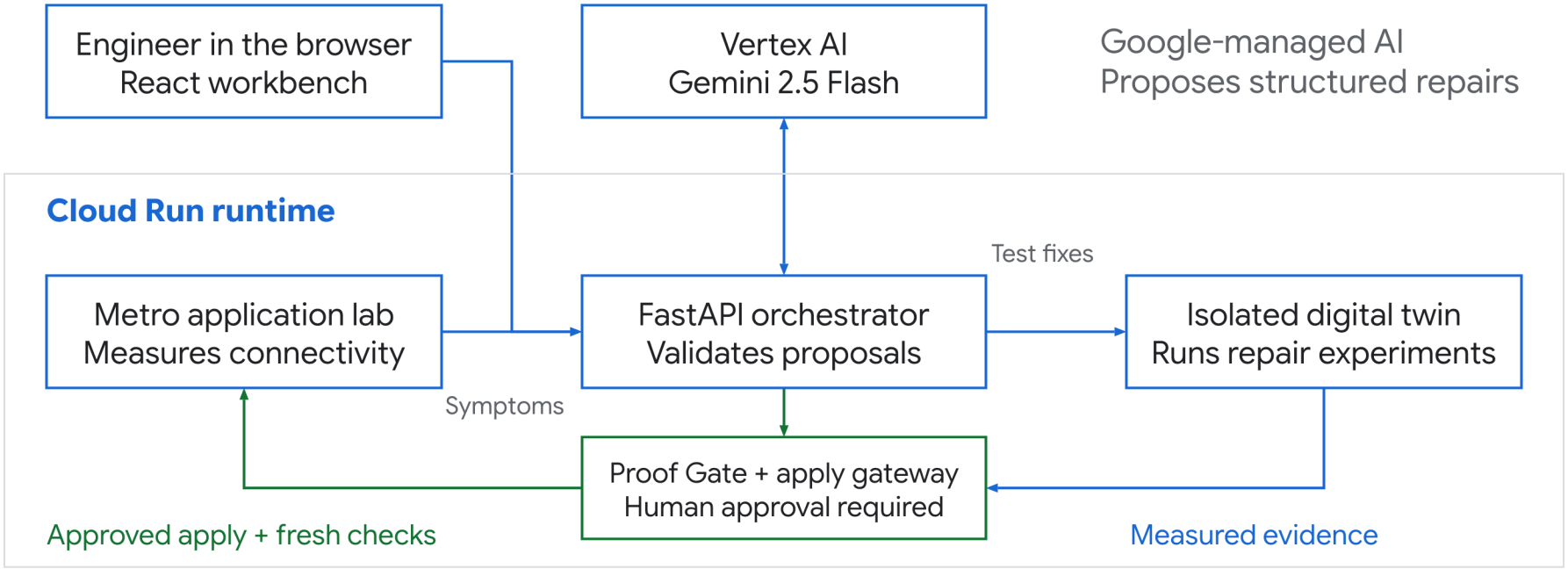
Rehearse reversal before approval.

Visible limits

Show what the twin did not test.

Recorded workbench, 15 October 2026. Passing tests alone does not authorize a network change.

Architecture diagram of the proposed solution



FRR / CoreDNS / nginx run the protocols. SQLite stores evidence on the instance.
Secret Manager supplies credentials. Cloud Build + Artifact Registry deliver the container.

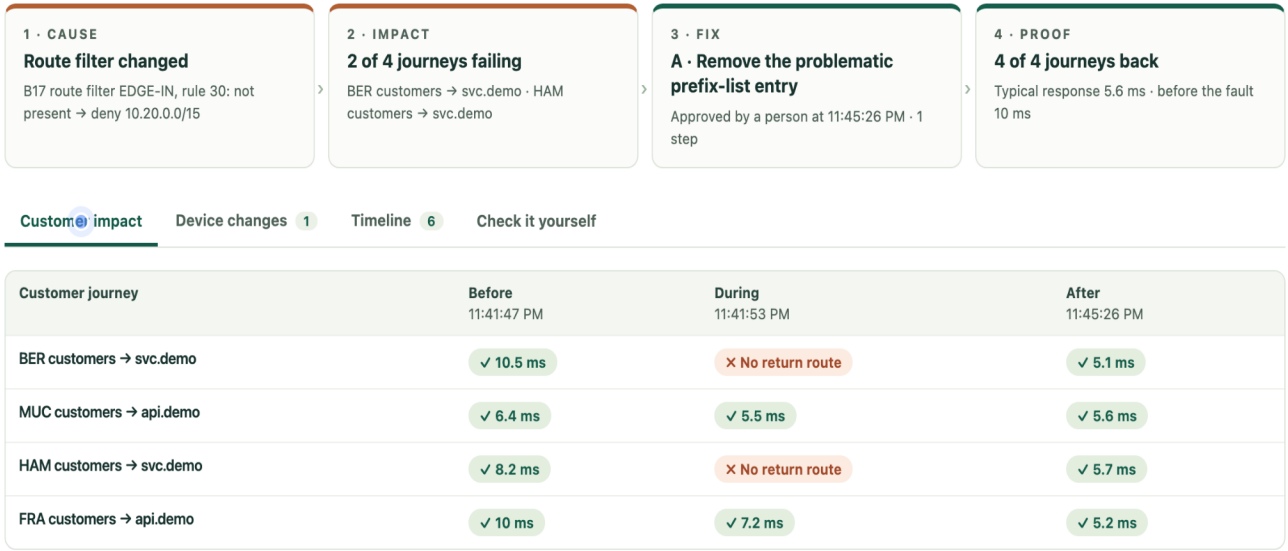
Technologies used in the solution

Technology	Role in the working prototype
Cloud Run (second generation)	Hosts TwinOps, Metro and isolated Linux namespace networks
Gemini 2.5 Flash / Vertex AI	Hypotheses, repair proposals, challenges and bounded revisions
Google Gen AI SDK	Structured model requests with schema and catalogue validation
Secret Manager	Stores the server credential for protected approval mode
Cloud Build / Artifact Registry	Builds and stores the deployment container
React / FastAPI / SQLite	Review interface, orchestration and instance-local evidence
FRR / CoreDNS / nginx	Real routing, DNS, IPv4/IPv6 and TLS experiments

Firebase and Firestore are not used. Cloud Run provides the deployment path.

Snapshots of the prototype

Two failed journeys. One rule removed. Four journeys healthy.



Independent check

Metro probes the network again.

Berlin and Hamburg recover. Munich and Frankfurt stay healthy.

The planner does not score its own success.

Actual hosted run, 15 October 2026. Synthetic sites and customer counts. Timings are illustrative.

Prototype performance report / benchmarking

Evidence scope	Observed result
Hosted Gemini repair 15 October 2026	10 model calls, 0 fallbacks and 0 diagnosis writes
Candidate experiments Same hosted repair	5 experiments, 13 challenges and 79 probes
Post-change recovery Same hosted repair	11/11 checks passed in each of 3 observation windows
Deterministic evaluation 13 October 2026	320 cases: 200 verified repairs, 120 abstentions, 0 unsafe passes; 13/13 controls passed

The 320-case suite uses a deterministic model twin and makes zero Gemini calls.
Production reliability and operator time savings still need separate evaluation.

Additional details / future development

Today: a shared prototype

Routing, DNS and TLS exercises on synthetic topology.
One lab instance, ephemeral evidence and open demo approval.

Next: operator evaluation

Compare investigation and review effort against a baseline.
Measure escalation, rollback outcomes and decision quality.

Then: a read-only shadow pilot

Validate network adapters and twin coverage against real environments before enabling operational change.

For wider deployment

Durable evidence storage, individual identity, per-target isolation, owned monitoring and tested recovery are required.

Public repository, demo video and final product

GitHub public repository

https://github.com/prabhakarcs786/twinops

Demo video

https://youtu.be/0LjDHiUBy0Y

Final product: TwinOps workbench

https://twinops-live-705406130795.europe-west1.run.app/workbench

Connected application: Metro Network

https://twinops-live-705406130795.europe-west1.run.app/metro/

Review access and walkthrough: GitHub / docs / JUDGE_GUIDE.md

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Thank you

